

Please Review yourself the following topics in vector calculus; (EE1203)

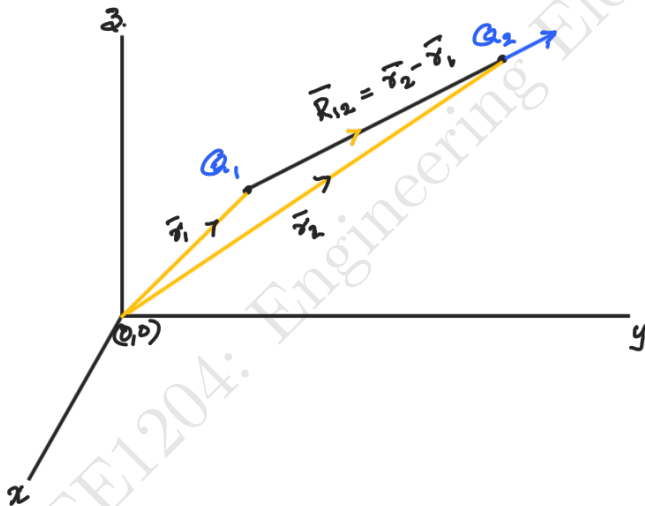
1. Vector algebra : vector addition, dot product, cross product.
2. Del operator : Gradient, curl, Divergence of vector fields.
3. Curvilinear Co-ordinates: Cylindrical & Spherical.
4. Integral Calculus: Line, surface & volume integrals, Divergence theorem, Stokes's theorem, Green's theorem.

Part 1: Electrostatics:

① Coulomb's Law and Electric Field Intensity.

Coulomb's law

* Experimental Law



Force acting on Q_2 by Q_1

$$|\vec{F}_{12}| \propto \frac{Q_1 Q_2}{|\vec{R}_{12}|^2}$$

$$\hat{F}_{12} = \hat{a}_{12} ;$$

$$\vec{F}_{12} = K \cdot \frac{Q_1 Q_2}{|\vec{R}_{12}|^2} \hat{a}_{12}$$

Unit vectors

$$\hat{a}_{12} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|} = \frac{\vec{r}_2 - \vec{r}_1}{|\vec{r}_2 - \vec{r}_1|}$$

Force on Q_1 by Q_2

$$\vec{F}_{21} = K \cdot \frac{Q_2 Q_1}{|\vec{R}_{21}|^2} \hat{a}_{21}$$

$$\vec{R}_{21} = \vec{r}_1 - \vec{r}_2 ; \hat{a}_{21} = \frac{\vec{R}_{21}}{|\vec{R}_{21}|}$$



In SI unit proportionality
constant $k = \frac{1}{4\pi\epsilon_0}$;

ϵ_0 = permittivity of Free space
 $\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$

Coulomb's law;

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\vec{R}_{12}|^2} \hat{q}_{12} ;$$

How powerful this electric force (Coulomb's force) is?

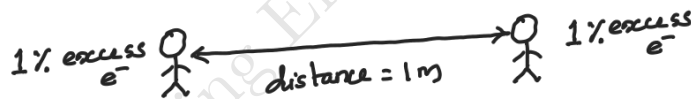
Ref: Lectures By Richard Feynman. Vol 2.

Example₁: → A human body which has +vely charged protons, ions & -vely charged electrons.

→ Assume an average human body weight of 100 kg

→ Human body density $\sim 1000 \text{ kg/m}^3$; $10^{23} \text{ e}^-/\text{cm}^3$
 $\sim 1 \text{ g/cm}^3$
 $\sim$ electron density

Assume human body has 1% excess e^-



1% excess $\text{e}^- \Rightarrow$ 1 kg of excess e^- in the body.

$$\Rightarrow \frac{1 \text{ kg}}{1000 \text{ kg/m}^3} \times \frac{10^{23} \text{ e}^-}{\text{cm}^3}$$

$$\frac{10^{23}}{\text{cm}^3} \approx \frac{10^{23} \times 10^6}{\text{m}^3}$$

$$\Rightarrow \approx \frac{1}{1000} \times 10^{23} \times 10^6 \text{ e}^- \approx 10^{26} \text{ e}^-$$

Electric Force between two human bodies with 1% excess e^-

$$|\vec{F}|_{\text{Elec}} = \frac{1}{4\pi\epsilon_0} \times \frac{10^{26} \text{ e}^- \times 10^{26} \text{ e}^-}{1^2} \approx 2.4 \times 10^{24} \text{ N}$$

$$\therefore \text{e}^- = 1.6 \times 10^{-19} \text{ C}$$

$$F = mg ; g \sim 10 \text{ m/s}^2$$

$$\Rightarrow m \sim 10^{23} \text{ kg}$$



This force can lift $\approx 10^{23}$ kg.

The mass of earth $\approx 10^{24}$ kg.

$\Rightarrow$ Just 1% excess e^- in our body (100 kg); the electric force experienced between two such body 1m apart could probably lift the earth.

Example 2: According to gravitational force two bodies attracts. But then why two e^- repels each other?

$e^- \quad d \quad e^-$: Force due to gravity

$$|\vec{F}_{\text{gravity}}| = \frac{G M m}{d^2} \\ = \frac{G m_e^2}{d^2}$$

Force due to electric force

$$|\vec{F}_{\text{electric}}| = \frac{1}{4\pi\epsilon_0} \frac{e^2}{d^2}$$

$$\frac{|\vec{F}_{\text{electric}}|}{|\vec{F}_{\text{gravity}}|} = \frac{e^2 / 4\pi\epsilon_0}{G m_e^2}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$\epsilon_0 = 8.85 \times 10^{-12} \frac{\text{C}^2}{\text{Nm}^2}$$

$$G = 6.7 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$= \frac{(1.6 \times 10^{-19})^2 / 4\pi\epsilon_0}{6.7 \times 10^{-11} \times (9.1 \times 10^{-31})^2} \\ \approx 10^{43}$$



How about slightly heavier particle ; protons?

Mass of proton $m_p \sim 2000 m_e$

|Charge of proton| = |Charge of e^- | = 1.6×10^{-19} .

⇒ For proton;

$$\frac{|\vec{F}_{\text{electric}}|}{|\vec{F}_{\text{gravity}}|} \approx \frac{10^{43}}{(2000)^2} \approx 10^{37}$$

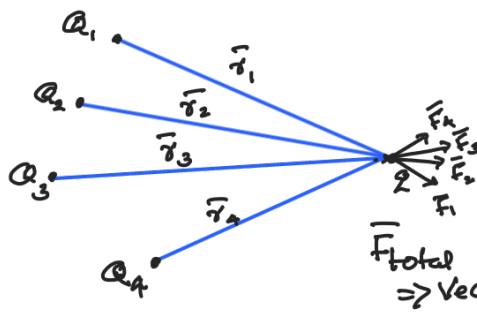
$$|\vec{F}_{\text{electric}}|_{\text{proton}} \approx 10^{37} |\vec{F}_{\text{gravity}}|_{\text{proton}}.$$

So Feynman's says;

Force due to electricity is "billion. billion. billion. billion" times stronger than gravity.
(1 billion $\approx 10^9$)

Superposition principle:

Suppose we have collection of charges (source) and need to calculate force on a test charge.



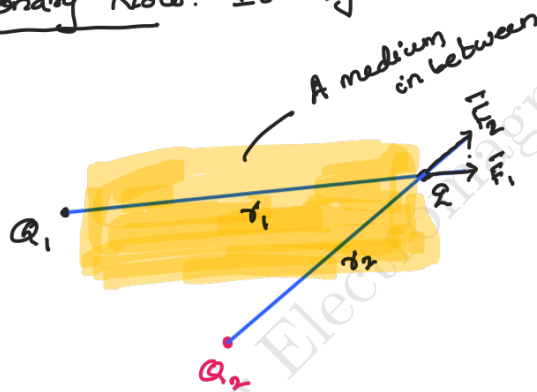
$$\begin{aligned} \vec{F}_{\text{total}} &= \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 \\ &= \frac{1}{4\pi\epsilon_0} \frac{Q_1 q}{r_1^2} \hat{r}_1 + \frac{1}{4\pi\epsilon_0} \frac{Q_2 q}{r_2^2} \hat{r}_2 \\ &\quad + \frac{1}{4\pi\epsilon_0} \frac{Q_3 q}{r_3^2} \hat{r}_3 + \frac{1}{4\pi\epsilon_0} \frac{Q_4 q}{r_4^2} \hat{r}_4 \end{aligned}$$

$$\vec{F}_{\text{total}} = \frac{1}{4\pi\epsilon_0} \left[\frac{Q_1}{r_1^2} \hat{r}_1 + \frac{Q_2}{r_2^2} \hat{r}_2 + \frac{Q_3}{r_3^2} \hat{r}_3 + \frac{Q_4}{r_4^2} \hat{r}_4 \right]$$

$$\boxed{\vec{F}_{\text{total}} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^4 \frac{Q_i}{r_i^2} \hat{r}_i} \quad \text{----- Superposition principle.}$$

Superposition principle says: Force between Q_i & Q_j is not affected by the presence of Q_k ($j \neq i$)

Precautionary Note: It might not be true always.



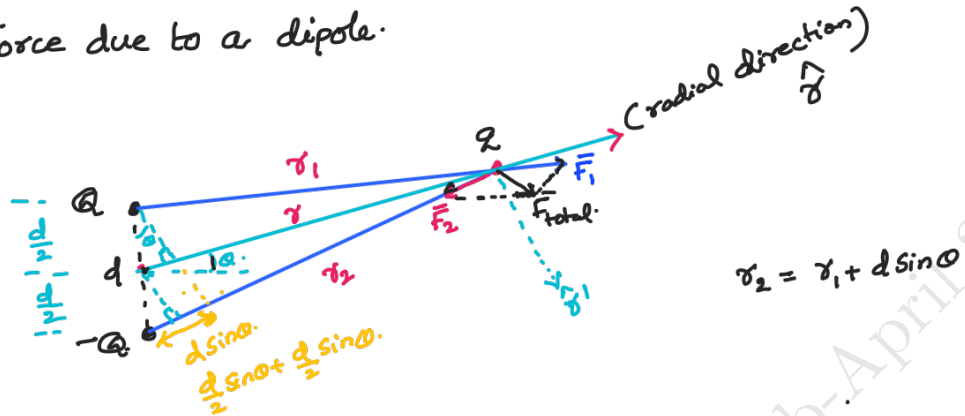
* Suppose the medium properties change due to Q_2 (another charge)

Example Suppose permittivity of medium is function of Q_2 .

$\Rightarrow$ Force (Coulomb's) changes in the presence of Q_2 as compared to that of absence of Q_2 . $\Rightarrow$ Superposition fails.

* But the experimental evidence proved that in vacuum/freespace superposition gives high accurate results

Example: Force due to a dipole.



$$\vec{F}_{\text{total}} = \vec{F}_1 + \vec{F}_2$$

$$= \left[\frac{1}{4\pi\epsilon_0} \frac{Qq}{r_1^2} - \frac{1}{4\pi\epsilon_0} \frac{Qq}{r_2^2} \right] \hat{r}_{\text{total}}$$

along radial
direction F_r

$$F_r = \frac{Qq}{4\pi\epsilon_0} \left[\frac{1}{r_1^2} - \frac{1}{(r_1 + d \sin \theta)^2} \right]$$

$$= \frac{Qq}{4\pi\epsilon_0} \left[\frac{1}{r_1^2} - \frac{1}{r_1^2 + 2dr_1 \sin \theta} \right] \quad d \ll r_1, r_2$$

$$= \frac{Qq}{4\pi\epsilon_0} \frac{1}{r_1^2} \left[1 - \frac{1}{\left(1 + \frac{2d}{r_1} \sin \theta\right)} \right]$$

$$= \underbrace{\frac{Qq}{4\pi\epsilon_0} \frac{1}{r_1^2}}_{F_1} \left[1 - \left[1 + \frac{2d}{r_1} \sin \theta \right]^{-1} \right] \quad \left(1 + \frac{2d}{r_1} \sin \theta\right)^{-1} \approx 1 - \frac{2d}{r_1} \sin \theta$$

$$= F_1 \left[\frac{2d \sin \theta}{r_1} \right] \quad \therefore d \ll r_1$$

* Similarly need to find $\vec{F}_\theta$! (Try yourself & find $\vec{F}_{\text{total}}$)

* Interpret the result when $d \rightarrow 0$

